

## **15 AVALON ROAD, AVALON**

### *Hydrological Engineering Scope of Works (Phase 3 – 5)*

29 May 2026

## **1. Project Background**

The 140-hectare industrial subdivision at 15 Avalon Road, Avalon sits within the Greater Avalon Employment Precinct (GAEP), subject to Development Plan Overlay 50 (*draft* DPO50, attached). Phase 1 and Phase 2 hydrological works have been completed by RAIN Consulting (provided later), establishing the catchment analysis and preliminary concept design. The DPO50 panel hearing process is complete, with gazettal of the amendment to follow. This scope addresses the remaining hydrological engineering required to deliver the water-related plans under Clause 4.0 of DPO50.

## **2. Scope of Works**

### **Phase 3: DPO50 Clause 4.0 Water-Related Plans**

Preparation of the three coordinated water-related documents required under DPO50 Clause 4.0 (refer attached DPO50 for full requirements):

#### ***Phase 3.1: Drainage and Stormwater Management Strategy (CCMA and CoGG approval)***

Strategy report addressing all matters listed under DPO50 Clause 4.0 for the Drainage and Stormwater Management Strategy (DSMS). Includes updated RORB and MUSIC modelling, Flood Impact and Risk Assessment (1% AEP under future climate), review of the Ramsar Wetlands and Avalon Coastal Reserve environmental impact assessment, and concept design of basins, outfalls, and overland flow paths. The scope of works should include:

- Review of existing strategies and background reports.
- Undertake GAP analysis and review available LiDAR, survey, aerial imagery, and existing infrastructure (as available).
- Update Catchment Plan.
- Review and update existing Rain hydrological model (RORB) for existing and developed conditions, including the location and sizing of retarding basins.
- Prepare a Flood Impact Assessment hydraulic model (TUFLOW). Models are required for existing and developed condition. Allowance for climate change per DPO.
- Update the MUSIC model for developed conditions to ensure the proposed stormwater treatment train is fit for purpose and meets best practice, including the location and size of water sensitive urban design (WSUD) assets.

- Liaise with Engineering Consultant to co-ordinate key utility servicing and development sequencing.
- Develop conceptual plan for proposed drainage/waterway corridors and WSUD assets, including retarding basins, sediment ponds and wetlands.
- Document all calculations, parameters and outcomes into the DSMS Report.

### *Phase 3.2: Drainage Asset Management Plan (CoGG approval)*

Drainage Asset Management Plan addressing the requirements of DPO50 Clause 4.0 for drainage assets. Includes asset register, public/private ownership delineation, long-term operational and maintenance arrangements, agreed maintenance schedule and cost apportionment, lifecycle costing, and inspection/monitoring protocols.

### *Phase 3.3: Integrated Water Management Plan (CoGG approval)*

IWMP addressing the requirements of DPO50 Clause 4.0 for the integrated water management system. Covers the full water cycle including stormwater treatment/storage/reuse (consistent with Phase 3.1), wastewater reuse potential, potable water offset, stormwater harvesting feasibility, and long-term ownership of reuse infrastructure.

The Environmental Impact Assessment (EIA) of the Ramsar Wetlands and Avalon Coastal Reserve has not commenced. This EIA will inform stormwater reduction targets when provided. Therefore, the appointed consultant is to provide a range of stormwater volume reduction options for consideration and implementation.

Options will need to be workshopped with Barwon Water, CoGG and the CCMA (and any other relevant stakeholders). The consultant is to make allowance for a minimum of three workshops.

The scope of works should include:

- Review any background studies relating to potable water reduction and stormwater volume reduction initiatives relative to the site.
- Establish stormwater volume reduction targets (note this is to be at least partially informed by the EIA).
- Qualitative analysis of IWM “long list” through the lens of volume reduction, considering lot scale, precinct scale, and regional scale infrastructure.
- Undertake MUSIC modelling to quantify predicted volume reduction for each of the shortlist IWM options.
- Refine the “long list” to a “short list” in consultation with the Landowner Group and key stakeholders.
- Determine a preferred portfolio of options which maximises IWM objectives and reaches the volume reduction target.
- Prepare a high-level concept plan presenting the selected IWM options.

- Include an IWM section in the DSMS Report, including a summary of IWM options selection and volume reduction analysis.
- Allowance for 3 x 2hr stakeholder workshop meetings to identify and present options.

#### Phase 4: Functional Design Plans and Report (CoGG approval)

Development of concept designs to functional design standard in accordance with CoGG requirements. Includes detailed hydraulic modelling, functional design of drainage infrastructure, basin design, integration with upstream/downstream systems, and a Functional Design Report with CAD drawings suitable for CoGG approval.

#### Phase 5.1: Asset Costing (Provisional)

Preparation of a detailed asset cost estimate covering the drainage and stormwater infrastructure designed under Phase 4, to support landowner contribution apportionment, ICP/DCP submissions and reimbursement claims with the relevant drainage authority. The fee for this phase is provisional and to be calculated as a percentage of the estimated construction cost in accordance with the ACEA sliding scale.

#### Phase 5.2: Ongoing Technical Advice

Ongoing technical support on an as-required basis. Includes response to referral authority comments and conditions (CCMA, CoGG), technical submissions responses, resolution of issues, interdisciplinary coordination, and supplementary studies as identified by authorities. Additional scope beyond this allowance will be quoted separately.

### 3. Deliverables, Approving Authority & Fee Basis

| Phase | Deliverable                                 | Approving Authority | Fee Basis                  |
|-------|---------------------------------------------|---------------------|----------------------------|
| 3.1   | Drainage and Stormwater Management Strategy | CCMA & CoGG         | Fixed fee                  |
| 3.2   | Drainage Asset Management Plan              | CoGG                | Fixed fee                  |
| 3.3   | Integrated Water Management Plan            | CoGG                | Fixed fee                  |
| 4     | Functional Design Plans & Report            | CoGG                | % Fee (ACEA sliding scale) |
| 5.1   | Asset Costing (Provisional)                 | N/A                 | Fixed fee                  |
| 5.2   | Ongoing Technical Advice (as required)      | N/A                 | T&M                        |

The client will provide RAIN Consulting's Phase 1 & 2 reports. These reports consist of RORB, MUSIC and Catchment plans in GIS format.

Please advise if you require any clarifications.